

Chapter

5

Normal Probability Distributions

GLOBAL
EDITION



Elementary Statistics

Picturing the World

EIGHTH EDITION

Ron Larson



Chapter Outline

- 5.1 Introduction to Normal Distributions and the Standard Normal Distribution
- 5.2 Normal Distributions: Finding Probabilities
- 5.3 Normal Distributions: Finding Values
- 5.4 Sampling Distributions and the Central Limit Theorem
- 5.5 Normal Approximations to Binomial Distributions

Section 5.3

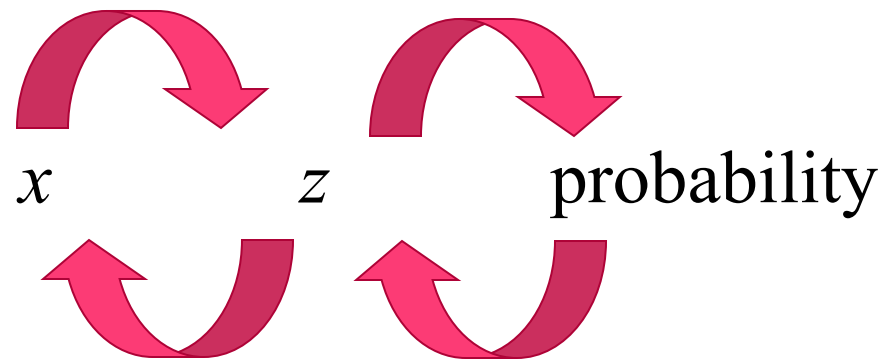
Normal Distributions: Finding Values

Section 5.3 Objectives

- How to find a z -score given the area under the normal curve
- How to transform a z -score to an x -value
- How to find a specific data value of a normal distribution given the probability

Finding values Given a Probability

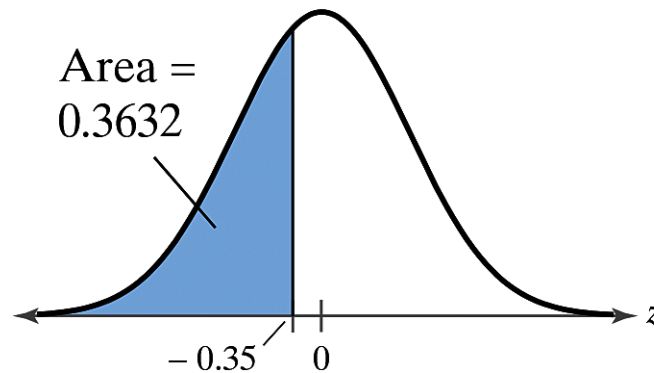
- In a previous section we were given a normally distributed random variable x and we were asked to find a probability.
- In this section, **we will be given a probability and we will be asked to find the value of the random variable x .**



Example: Finding a z-Score Given an Area

1. Find the z-score that corresponds to a cumulative area of 0.3632.

Solution:



Solution: Finding a z-Score Given an Area

- Locate 0.3632 in the body of the Standard Normal Table.

z	.09	.08	.07	.06	.05	.04	.03
−3.4	.0002	.0003	.0003	.0003	.0003	.0003	.0003
−0.5	.2776	.2810	.2843	.2877	.2912	.2946	.2981
−0.4	.3121	.3156	.3192	.3228	.3264	.3300	.3336
−0.3	.3483	.3520	.3557	.3594	.3632	.3669	.3707
−0.2	.3859	.3897	.3936	.3974	.4013	.4052	.4090

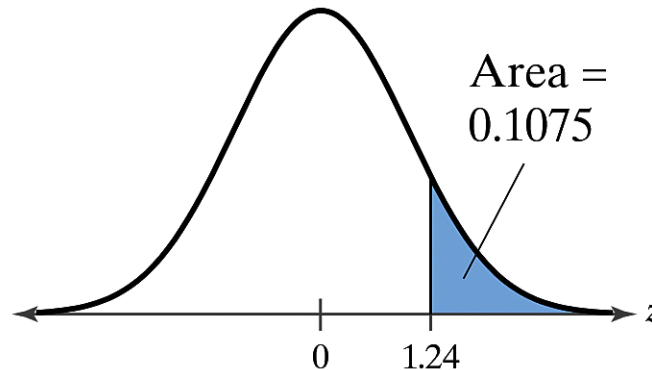
- The values at the beginning of the corresponding row and at the top of the column give the z-score.

The z-score is − 0.35.

Example: Finding a z-Score Given an Area

2. Find the z-score that has 10.75% of the distribution's area to its right.

Solution:



Because the area to the right is 0.1075, the cumulative area is $1 - 0.1075 = 0.8925$.

Solution: Finding a z-Score Given an Area

- Locate 0.8925 in the body of the Standard Normal Table.

z	.00	.01	.02	.03	.04	.05	.06
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131

- The values at the beginning of the corresponding row and at the top of the column give the z-score.

The z-score is 1.24.

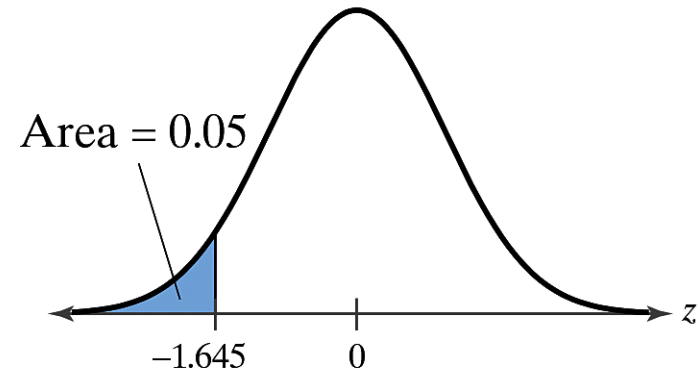
Example: Finding a z-Score Given a Percentile

Find the z-score that corresponds to each percentile.

1. P_5 .

Solution:

The z-score that corresponds to P_5 is the same z-score that corresponds to an area of 0.05.



The areas closest to 0.05 in the table are 0.0495 ($z = -1.65$) and 0.0505 ($z = -1.64$). Because 0.05 is halfway between the two areas in the table, use the z-score that is halfway between -1.64 and -1.65 . **The z-score is -1.645.**

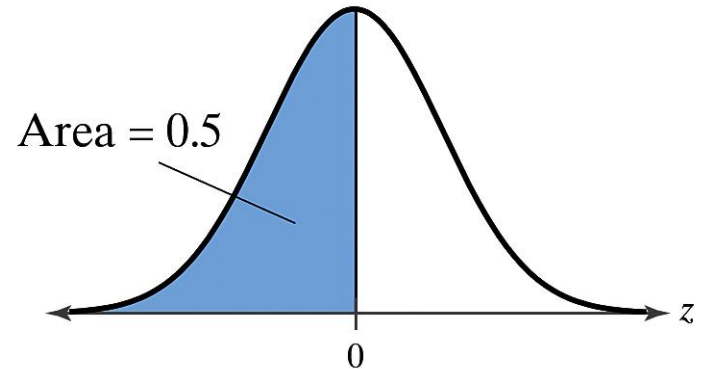
Example: Finding a z-Score Given a Percentile

Find the z-score that corresponds to each percentile.

2. P_{50} .

Solution:

The z-score that corresponds to P_{50} is the same z-score that corresponds to an area of 0.50.



Locate 0.5 in the Standard Normal Table. The area closest to 0.5 in the table is 0.5000, so **the z-score** that corresponds to an area of 0.5 **is 0**.

Example: Finding a z-Score Given a Percentile

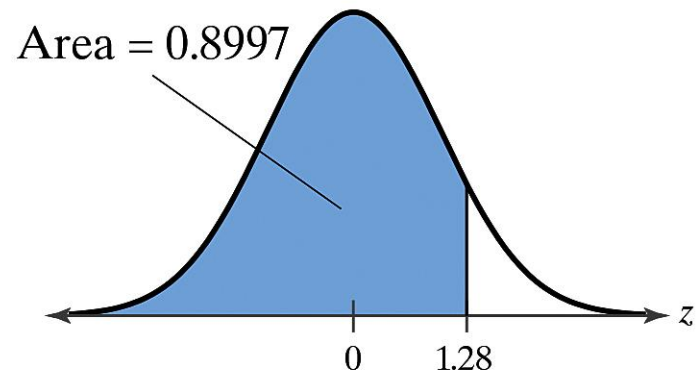
Find the z-score that corresponds to each percentile.

3. P_{90} .

Solution:

The z-score that corresponds to P_{90} is the same z-score that corresponds to an area of 0.90.

Locate 0.9 in the Standard Normal Table. The area closest to 0.9 in the table is 0.8997, so **the z-score** that corresponds to an area of 0.9 **is about 1.28**.



Example: Finding a z-Score Given a Percentile

Solution:

You can use technology to find the z-score that corresponds to each percentile, as shown below.

Remember that when you use technology, your answers may differ slightly from those found using the Standard Normal Table.

EXCEL		
	A	B
1	1.NORM.INV(0.05,0,1)	
2		-1.644853627
3	2.NORM.INV(0.50,0,1)	
4		0
5	3.NORM.INV(0.90,0,1)	
6		1.281551566

Transforming a z-Score to an x -Value

To transform a standard z -score to a data value x in a given population, use the formula

$$x = \mu + z\sigma$$

Example: Finding an x -Value Corresponding to a z -score

A veterinarian records the weights of cats treated at a clinic. The weights are normally distributed, with a mean of 9 pounds and a standard deviation of 2 pounds. Find the weights x corresponding to each z -score. Interpret the results.

1. $z = 1.96$ 2. $z = -0.44$ 3. $z = 0$.

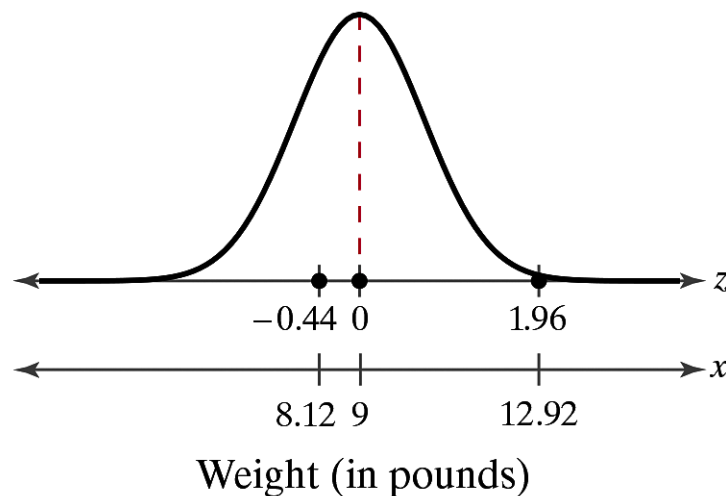
Solution: Use the formula $x = \mu + z\sigma$

1. $z = 1.96$: $x = 9 + 1.96(2) = 12.92$ pounds
2. $z = -0.44$: $x = 9 + (-0.44)(2) = 8.12$ pounds
3. $z = 0$: $x = 9 + 1.96(0) = 9$ pounds

Example: Finding an x -Value Corresponding to a z -score

Solution:

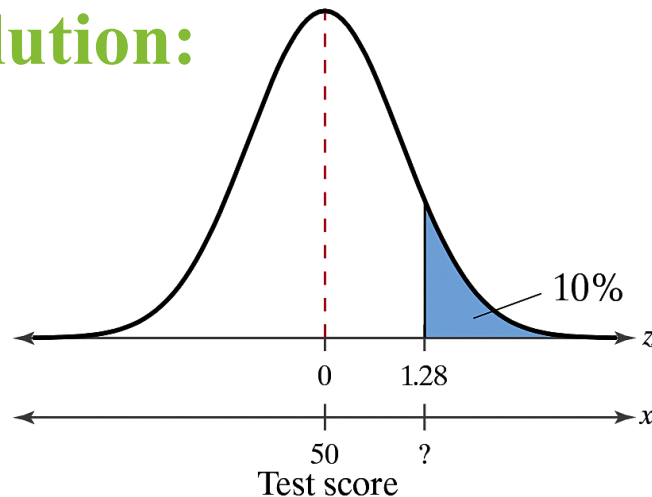
From the figure, you can see that 12.92 pounds is to the right of the mean, 8.12 pounds is to the left of the mean, and 9 pounds is equal to the mean.



Example: Finding a Specific Data Value for a Given Probability

Scores for the California Peace Officer Standards and Training test are normally distributed, with a mean of 50 and a standard deviation of 10. An agency will only hire applicants with scores in the top 10%. What is the lowest score you can earn and still be eligible to be hired by the agency? (*Source: State of California*)

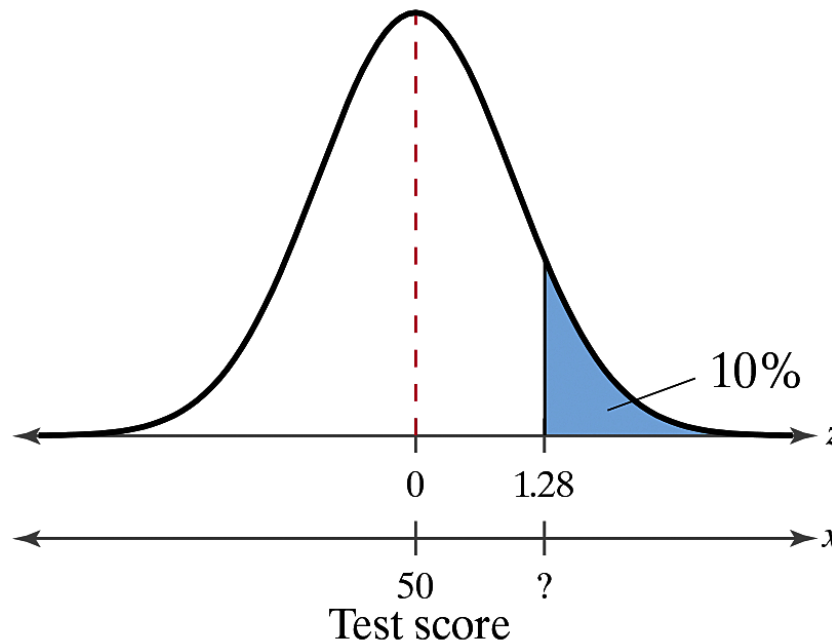
Solution:



An exam score in the top 10% is any score above the 90th percentile. Find the z-score that corresponds to a cumulative area of 0.9.

Solution: Finding a Specific Data Value for a Given Probability

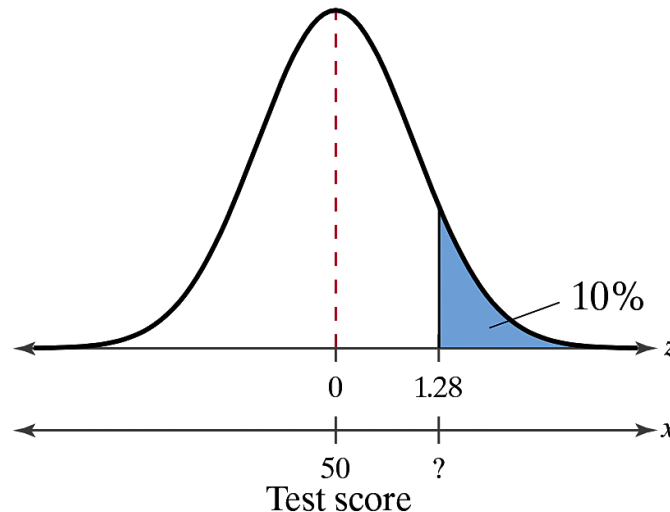
From the Standard Normal Table, the area closest to 0.9 is 0.8997. So the z -score that corresponds to an area of 0.9 is $z = 1.28$.



Solution: Finding a Specific Data Value for a Given Probability

Using the equation $x = \mu + z\sigma$

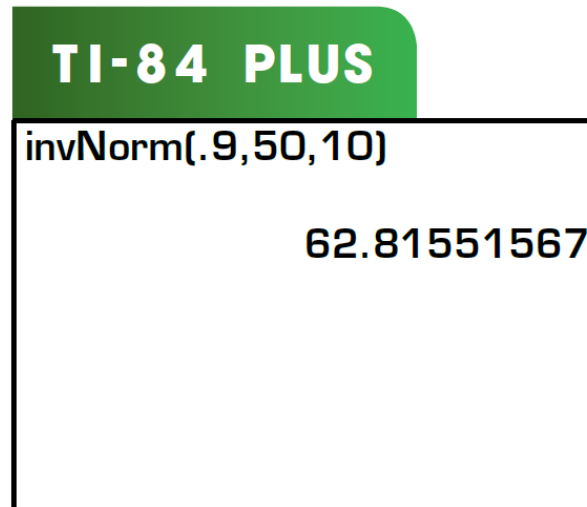
$$x = 50 + 1.28(10) \approx 62.8$$



The lowest score you can earn and still be eligible to be hired by the agency is about 63.

Solution: Finding a Specific Data Value for a Given Probability

You can check this answer using technology. For instance, you can use a TI-84 Plus to find the x -value, as shown.



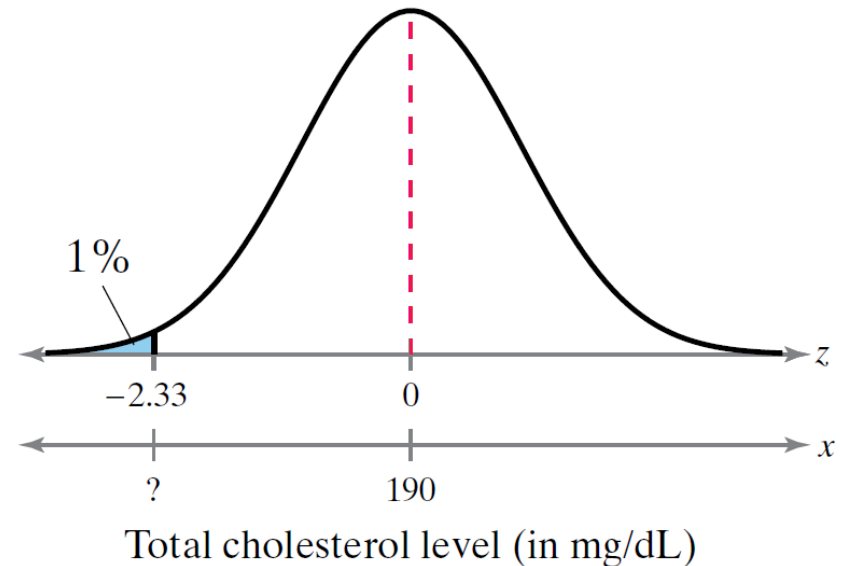
Example: Finding a Specific Data Value

In a randomly selected sample of U.S. adults ages 20 and over, the mean total cholesterol level is 190 milligrams per deciliter with a standard deviation of 40.9 milligrams per deciliter. Assume the total cholesterol levels are normally distributed. Find the highest total cholesterol level an adult aged 20 or over can have and still be in the bottom 1%. *(Adapted from National Center for Health Statistics)*

Solution: Finding a Specific Data Value

Solution:

Total cholesterol levels in the lowest 1% correspond to the shaded region shown.



In the Standard Normal Table, the area closest to 0.01 is 0.0099. So, the z -score that corresponds to an area of 0.01 is $z = -2.33$. To find the x -value, note that $\mu = 190$ and $\sigma = 40.9$, and use the formula $x = \mu + z\sigma$.

$$x = \mu + z\sigma = 190 + (-2.33)(40.9) \approx 94.7$$

Solution: Finding a Specific Data Value

Solution:

You can check this answer using technology. For instance, you can use Excel to find the x -value, as shown.

EXCEL	
	A
1	94.85237195 ← =NORM.INV(0.01,190,40.9)

The value that separates the lowest 1% of total cholesterol levels for U.S. adults ages 20 and over from the highest 99% is about 95 milligrams per deciliter.